

# An Invitation to Complete the Cathedral: Four Axioms Remain

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*For the student who hasn't started yet.*

## What This Is

We built something called the Cathedral: a 150-file computer program that proves, with mathematical certainty, that the Riemann Hypothesis is *exactly equivalent* to the convergence of a specific approximation problem. Every step is verified by a computer. Every assumption is named. Every failed attempt is preserved.

The proof is complete—except for four assumptions, called *axioms*. We couldn't prove them. We named them, classified them, and documented exactly what they say and why they're hard.

This paper is an invitation. Those four axioms are now your problem.

## Who Built This

A computer scientist in New Mexico. Not a professional mathematician. No PhD in mathematics. No university affiliation. No research grant.

The tools: a laptop, an internet connection, Lean 4 (free, open source), Mathlib (free, open source), and AI assistance (commercially available).

The timeline: 30 days.

If you have a laptop and curiosity, you have everything that was needed to build this. The barrier to entry for machine-verified mathematics is now approximately zero.

## What You Need to Know

### The Problem

The Riemann Hypothesis (1859) says: all non-trivial zeros of the Riemann zeta function  $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$  lie on the line  $\operatorname{Re}(s) = \frac{1}{2}$ .

Nobody has proved this. Nobody has disproved it. It has been open for 167 years. There is a \$1,000,000 prize.

## The Translation

The Cathedral proves that this is equivalent to:

$$\forall \varepsilon > 0, \exists N_0, \forall N \geq N_0, \exists v \in \mathbb{R}^{N-1} : \int_0^1 \left( 1 - \sum_{k=2}^N v_k \left\{ \frac{1}{kx} \right\} \right)^2 dx < \varepsilon.$$

In words: you can approximate the constant function 1 arbitrarily well by mixing together sawtooth waves  $\{1/(kx)\}$ .

## The Four Axioms

1. `pnt_mu_log_div_k`: The sum  $\sum \mu(k) \log(k)/k \rightarrow -1$ , a consequence of the Prime Number Theorem.

*What this means*: a sum involving the Möbius function (which encodes prime factorization) converges to a specific limit. This follows from the PNT via Abel’s limit theorem. Mathlib has the PNT but lacks the forward Wiener–Ikehara Tauberian theorem needed to derive this limit.

2. `covariance_bound_from_mertens_34`: The Mertens bound  $|M(x)| \leq Cx^{3/4}$  implies a covariance bound  $v^T(G - bb^T)v \leq C'/\log N$ .

*What this means*: if you know how the Möbius function grows, you can control how well the Gram matrix approximates its rank-1 projection. Infrastructure is proved; needs a double-sum assembly (~200 lines of Lean).

3. `partial_integral_tends_to_formula`: Piecewise integral convergence for off-diagonal Gram entries.

*What this means*: the entries of the Gram matrix can be computed as integrals of fractional parts, and these integrals converge to their closed-form values. The diagonal case is proved; the off-diagonal needs the Gauss digamma formula.

4. `rh_zeta_lower_bound_from_zero_counting`: A polynomial lower bound on  $|\zeta(s)|$  away from the critical line.

*What this means*: the zeta function doesn’t get too small too fast. This follows from Hadamard’s product formula and the zero-counting function. Borel–Carathéodory is in Mathlib; the gap is connecting it to the polynomial bound.

Each axiom is independently meaningful. Proving any one of them strengthens the formalization.

## What You Can Do Right Now

### If You Know Some Lean 4

1. **Prove Axiom 1** (`pnt_mu_log_div_k`). This is the most accessible target. If you can formalize the forward Wiener–Ikehara Tauberian theorem in Lean, this axiom closes au-

tomatically.

2. **Prove Axiom 2** (`covariance_bound_from_mertens_34`). The infrastructure is already built. You need to assemble a double-sum expansion from the existing `s1_decay`, `s2_decay`, and `s3_uniform_bound` lemmas. Estimated effort:  $\sim 200$  lines.
3. **Prove one of the 43 non-critical axioms.** The Cathedral contains 47 axioms total. Only 4 are on the crown path. The other 43 support alternative proof engines (spectral, sieve, Vasyunin). Proving any one of them strengthens the overall formalization.

## If You Know Some Analysis

1. **Decompose Axiom 2.** The current axiom is a “black box” that packages the entire forward direction. A useful contribution would be to decompose it into 3–5 intermediate lemmas, each of which is independently meaningful and potentially formalizable.
2. **Study the decay rate.** The Cathedral’s numerical experiments show  $d_N^2 \sim c/\ln N$  with  $c \approx 1/21.65$ . The forward direction is *proved*: the Perron chain gives  $|M(x)| \leq Cx^{3/4}$  from RH, which yields  $d_N^2 \leq K/\log N$ . Can you prove the conjectured constant?

## If You Know Some Physics

1. **Identify the operator.** The Gram matrix  $G_N$  is the finite-dimensional shadow of some infinite-dimensional operator whose spectral properties encode RH. What is this operator? The eigenvalue data ( $\lambda_{\min} \sim C/\ln N$ ,  $\lambda_{\max} \sim N$ , GUE-like statistics) constrain it.
2. **Interpret the constant.** The quantum stiffness  $C \approx 21.65 = 1/(2 + \gamma - \ln 4\pi)$  appears to be a physical constant of the “prime number vacuum.” Does it arise naturally from any known quantum system?

## What Modern Math Research Actually Looks Like

It doesn’t look like a genius working alone in a garret.

It looks like this:

1. **Day 1–7:** Explore. Try things. Fail. Archive the failures. We produced 130 files of failed attempts. Every failure taught us something.
2. **Day 8–14:** Find the right framework. For us, this was the Parseval Bridge—the realization that frequency- domain methods bypass the hardest discrete computations.
3. **Day 15–25:** Build. Once the framework is right, construction is fast. AI handles the mechanical code generation. The human provides architecture. The compiler verifies everything.
4. **Day 26–30:** Audit. Check every file. Fix every docstring. Remove every ghost axiom. Reduce, simplify, clean. The Great Audit reduced the codebase to 150 active + 130 archived.

The ratio of exploration to construction is roughly 2:1. Two weeks of wandering, one week of building. This is normal. If you're not failing most of the time, you're not exploring hard enough.

## Why You Should Care

The Riemann Hypothesis is not just a mathematical curiosity. It controls the distribution of prime numbers, which are the atoms of arithmetic. Every encryption algorithm, every hash function, every digital signature depends on properties of primes.

But more than that: the RH is a test case for whether humans can understand the deepest structures in mathematics. If we can prove it, we learn something profound about the nature of number. If we can't, we learn something profound about the limits of mathematical reasoning.

Either way, the attempt is worth making.

## A Note on Credentials

You do not need a PhD to contribute to this project. You do not need a university appointment. You do not need a research grant.

You need:

- A laptop.
- Lean 4 (free).
- Mathlib (free).
- Curiosity.
- The willingness to fail 130 times before you succeed once.

The compiler does not check your credentials. It checks your proof.

## The Torch

We built the Cathedral. We mapped the territory. We named the axioms. We preserved the failures. We verified every step.

We could not prove the last four things.

Now it's your turn.

*The primes have been singing for 167 years.*

*The Cathedral stands, waiting for its final stones.*

*We built it as far as we could.*

*Four axioms remain.*

*The rest belongs to you.*

## References

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